

Math 142, S14

Quiz 5

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Determine whether the following improper integrals are convergent or divergent and evaluate those that are convergent.

$$(a) \int_2^{+\infty} e^{-x/2} dx$$

$$\begin{aligned} & \int_2^{+\infty} e^{-x/2} dx \\ &= \lim_{t \rightarrow +\infty} \int_2^t e^{-x/2} dx \\ &= \lim_{t \rightarrow +\infty} [-2e^{-x/2}]_{x=2}^{x=t} \\ &= \lim_{t \rightarrow +\infty} -2(e^{-t/2} - e^{-1}) \\ &= -2(0 - e^{-1}) \\ &= \frac{2}{e} \end{aligned}$$

$$(b) \int_{-\infty}^{2\pi} \cos \theta d\theta$$

$$\begin{aligned} & \int_{-\infty}^{2\pi} \cos \theta d\theta \\ &= \lim_{t \rightarrow -\infty} \int_t^{2\pi} \cos \theta d\theta \\ &= \lim_{t \rightarrow -\infty} [\sin \theta]_{\theta=t}^{\theta=2\pi} \\ &= \lim_{t \rightarrow -\infty} (\sin 2\pi - \sin t) \\ &= -\lim_{t \rightarrow -\infty} \sin t \rightarrow \text{the limit doesn't exist} \\ &\Rightarrow \text{divergent.} \end{aligned}$$

Problem 2. (4 points) Evaluate the improper integral $\int_2^3 \frac{1}{\sqrt{x-2}} dx$ if possible.

$\frac{1}{\sqrt{x-2}}$ is discontinuous at $x=2$. Thus.

$$\begin{aligned} \int_2^3 \frac{1}{\sqrt{x-2}} dx &= \lim_{t \rightarrow 2^+} \int_t^3 \frac{1}{\sqrt{x-2}} dx \\ &= \lim_{t \rightarrow 2^+} [2\sqrt{x-2}]_{x=t}^{x=3} \\ &= \lim_{t \rightarrow 2^+} 2(1 - \sqrt{t-2}) \\ &= 2(1-0) = 2 \end{aligned}$$